Morgan Stanley | RESEARCH

UPDATE

October 27, 2025 11:25 PM GMT

Global Macro Commentary | Global

October 27: Trade Tensions Thaw

US & China signal progress towards trade agreement; US trade agreements in Southeast Asia; US extends tariff deadline for Mexico but proposes 10% increase on Canada; Milei's party wins Argentina midterms; US rates concession before front-end auctions; DXY at 98.78 (-0.2%); US 10y at 3.98 (-2.1bp).

MORGAN STANLEY & CO. LLC
Bradley Tian
Strategist
Bradley.Tian@morganstanley.com +1 212 761-0826
Molly Nickolin
Strategist
Molly.Nickolin@morganstanley.com +1 212 761-3592
Lingdi Xu
Economist
Lingdi.Xu@morganstanley.com +1 212 761-2957
Sofia Palacios
Strategist
Sofia.Palacios@morganstanley.com +1 212 761-0428

MORGAN STANLEY ASIA LIMITED+
Gek Teng Khoo
Strategist
Gek.Teng.Khoo@morganstanley.com +852 3963-0303

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Hiromu Uezato
Strategist
Hiromu.Uezato@morganstanleymufg.com +81 3 6836-8431

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+
Nimish M Prabhune
Strategist
Nimish.Prabhune@morganstanley.com +91 22 6996-1862

MORGAN STANLEY & CO. INTERNATIONAL PLC+
Maria Chiara Russo
Strategist
Maria.Chiara.Russo@morganstanley.com +44 20 7677-3499
David Cueva
Strategist
David.Cueva@morganstanley.com +44 20 7425-2143

Heading into a week of major central bank meetings, risk sentiment rebounds on the back of constructive signals from US-China trade talks over the weekend.

Developed Markets

- **Front-end Treasuries sell off in a concession ahead of the 2y and 5y auctions, while long-end yields drift lower (30y: -4bp) despite strong risk sentiment.** Both auctions stop close to the screws, resulting in front-end yields remaining elevated (2y: +1bp) through the afternoon. The long end outperforms, with the flattening move potentially supported by positioning on the curve into the front-end supply. Ahead of Wednesday's FOMC meeting, Treasury Secretary Bessent confirms five remaining candidates for the next Fed chair: Waller, Bowman, Warsh, Hassett, and Rieder. Labor market concerns remain in focus amid rumors of corporate layoffs and hiring slowdowns related to AI.

- **US equities rally 1.2% following positive readouts of the weekend's preliminary talks between US and Chinese officials.** President Trump says he "really feels good" about reaching a deal with Chinese President Xi when they meet on October 30. In line with our expectations for a limited US-China agreement, Treasury Secretary Bessent says the proposed 100% tariff is "off the table," potentially in exchange for China purchasing US soybeans and delaying its rare earth licensing regime "for a year." According to officials from both sides, other areas for discussion could include export controls, shipping fees, and fentanyl-related tariffs. Tech and other cyclical stocks (Communication Services: +2.3%; IT: +2.0%) lead the rally, while defensive sectors lag (Consumer Staples: -0.3%).

- **Risk-sensitive currencies exposed to trade with China lead gains among the G10 (AUD/USD: +0.7%; NZD/USD: +0.3%).** Safe-haven currencies underperform the move against the dollar, with CHF and JPY closing largely unchanged. Similarly, gold prices slide another 3% below the $4,000/oz threshold as safe-haven demand diminishes.

- **USD/CAD holds steady even as the US proposes additional 10% tariffs on Canada.** Market participants seemingly take into account the undefined implementation timeline and that most Canadian goods are exempt from US tariffs under the USMCA. Elsewhere in the G10, UK gilts bull-flatten (2y: -2bp; 30y: -4bp) and GBP outperforms (GBP/USD: +0.2%) on easing fiscal concerns as Chancellor Reeves maintains that tax hikes are a viable option to achieve fiscal goals in the Autumn Budget.

- **Duration rallies in the euro area, though front-end German bunds underperform (2y: +0bp) after the ifo Business Climate survey shows better-than-expected sentiment among German companies.** The flattening move is even more pronounced in EUR swaps, where 2s10s flattens by more than 2bp. EGB spreads tighten, particularly in Italy, where the 10y BTP-Bund spread narrows more than 1bp.

- **Japanese rates modestly bear-steepen (5y: +1bp; 30y: +3bp) on strong risk sentiment following the US-China trade talks, with the long end seemingly weighed down by political uncertainty ahead of this week's US-Japan summit.** In particular, market participants are focused on the possibility of increased defense spending as a potential policy concession to the US. Despite a rally in local equities (Nikkei: +2.5%), USD/JPY remains range-bound in light of the upcoming BoJ and FOMC meetings.

Emerging Markets

- **Asian rates broadly sell off on strong risk sentiment in the wake of the US-China trade talks.** However, Chinese rates rally, with CNY NDIRS yields falling 4bp after the PBoC signals that it will resume government bond trading operations. CNH strengthens 0.3% to 7.105 per USD, aided by a lower USD/CNY fixing (7.0881; P: 7.0928). Meanwhile, Korean rates bear-steepen (10y KRW NDIRS: +5bp) amid the risk-on rotation, with local equities rallying sharply (KOSPI: +2.6%) even as the US and Korea reiterate that a bilateral trade deal is unlikely to be finalized before Wednesday.

- **Malaysia and Thailand see currency strength and rates weakness after both countries sign trade agreements with the US.** In Malaysia, MYR rallies 0.3% and the 5y MYR NDIRS yield rises 4bp. In Thailand, rates also sell off (5y THB NDIRS: +3bp), but USD/THB holds steady at ~32.7 as trade optimism is offset by the aforementioned decline in gold prices. Meanwhile, INR and PHP both weaken 0.5% against USD amid unconfirmed reports of increased USD demand from Indian importers, as well as dovish comments from BSP in the Philippines (see below).

- **LatAm assets rally alongside the broader risk-on rotation across global markets, as well as constructive US trade developments in Brazil and Mexico.** BRL leads regional FX strength after the weekend meeting between Brazilian President Lula and US President Trump, described as positive by both parties. MXN also gains 0.3% against USD as Mexican President Sheinbaum announces that the US has extended its deadline for higher tariffs — originally set to take effect November 1 — by "several weeks" to allow for continued negotiations. In Chile, CLP's recent momentum stalls, closing flat against USD.

- **In Argentina, markets rally sharply after President Milei's party wins the midterm legislative election, likely providing enough votes to implement Milei's proposed fiscal reforms.** ARGENT'35 gains nearly 12 cash points, ARS gains 4% against USD, and local equities (MERVAL) rally 22%. Our economists believe the composition of the new government provides an opportunity to implement more sustained supply-side reforms.

- **Meanwhile, Colombia's curve sees mild bear steepening amid political noise surrounding the 2026 elections** and the initial candidate selection process, with turnout in the left-wing primaries exceeding historical levelsat nearly 3 million voters. Senator Cepeda securing 65% of the vote, signaling stronger-than-expected mobilization.

- **In a quiet session, CEEMEA currencies strengthen in line with the broader move against USD.** Hungary outperforms, with EUR/HUF closing the London session 0.4% lower. In Turkey, TRY weakens 0.1-0.2% against USD and EUR following a slight upside surprise in Turkey's unemployment rate (8.6%; P: 8.5%).

Central Bank Monitor

Emerging Markets

- In China, **PBoC Governor Pan** said the central bank will resume government bond trading operations, citing the current "sound" conditions of the overall bond market. He said the PBoC had paused bond trading earlier this year due to imbalance in bond supply and demand.
- In Philippines, **BSP Board Member Diokno** said he expects the central bank to lower its policy rate by another 25bp at its December meeting, with further cuts in 2026 possible depending on growth and employment data, as inflation is "pretty much under control." He reiterated the BSP does not target a specific exchange rate and only intervenes if it would affect the inflation target. He suggested the BSP should consider selling some of its gold holdings, which he views as excessive given that gold should ideally comprise 8-12% of the central bank's FX reserves. He revealed the BSP is still debating whether it should diversify the currencies in its FX reserves, possibly including EUR.

Exhibit 1: G4 Rates Closes (5pm NY)

	US			Germany			UK			Japan		
Tenor	27-Oct	1d Δ (bp)	2wk	27-Oct	1d Δ (bp)	2wk	27-Oct	1d Δ (bp)	2wk	27-Oct	1d Δ (bp)	2wk
2y	3.491	1.1		1.970	0.3		3.776	(1.6)		0.939	1.3	
3y	3.491	0.3		2.024	0.5		3.764	(1.7)		1.010	0.4	
5y	3.602	(0.4)		2.226	(0.4)		3.881	(1.9)		1.231	1.5	
7y	3.774	(1.3)		2.360	(0.7)		4.005	(2.8)		1.445	0.8	
10y	3.980	(2.1)		2.615	(1.0)		4.401	(3.0)		1.666	1.4	
20y	4.527	(3.5)		3.084	(1.8)		5.061	(3.4)		2.604	1.0	
30y	4.552	(4.1)		3.188	(1.7)		5.180	(3.8)		3.074	2.1	

Source: Morgan Stanley Research, Bloomberg

Exhibit 2: Inflation Closes (5pm NY)

	US TIPS			10y Real Yields, BEI								
Tenor	27-Oct	1d Δ (bp)	2wk	10y	27-Oct	1d Δ (bp)	2wk	10y	27-Oct	1d Δ (bp)	2wk
5y	1.220	1.5		DBRi	0.660	(1.0)		BTPi	1.712	(1.5)	
BEI	2.377	(2.0)		BEI	1.752	0.2		BEI	1.702	(0.8)	
10y	1.692	(0.9)		UKTi	1.549	(0.1)					
BEI	2.287	(1.2)		BEI	2.920	(3.1)					
30y	2.338	(2.9)		JGBi	0.047	0.5					
BEI	2.216	(1.1)		BEI	1.591	0.7					

Source: Morgan Stanley Research, Bloomberg

Exhibit 3: Macro Closes (5pm NY)

	Majors			EM				Equities / Commodities			
Spot	27-Oct	1d Δ (%)	2wk	Spot	27-Oct	1d Δ (%)	2wk	Index	27-Oct	1d Δ (%)	2wk
DXY	98.78	(0.2)		CNH	7.11	(0.2)		S&P	6875.16	1.2	
EUR	1.165	0.2		INR	88.24	0.4		Stoxx	5711.06	0.6	
GBP	1.334	0.2		THB	32.71	(0.2)		FTSE	9653.82	0.1	
JPY	152.88	0.0		KRW	1432.90	(0.4)		Nikkei	50512.32	2.5	
CHF	0.796	(0.0)		ZAR	17.22	(0.3)		VIX	15.79	(3.5)	
AUD	0.656	0.7		MXN	18.40	(0.3)		Gold (S)	3982.21	(3.2)	
CAD	1.399	(0.0)		BRL	5.37	(0.3)		WTI (Fut)	61.31	(0.3)	

Source: Morgan Stanley Research, Bloomberg

Exhibit 4: G4 Central Bank Implied Pricing

Central Bank OIS

Fed	Implied	1d Δ (bp)	Total Δ	ECB	Implied	1d Δ (bp)	Total Δ	BOE	Implied	1d Δ (bp)	Total Δ	BOJ	Implied	1d Δ (bp)	Total Δ
EFFR (10/24)	4.110	0.0	--	EONIA (10/24)	1.928	0.0	--	SONIA (10/24)	3.969	0.0	--	TONAR (10/24)	0.477	0.0	--
10/29/25	3.858	(0.5)	(25.2)	10/30/25	1.926	0.1	(0.2)	11/06/25	3.904	(0.7)	(6.5)	10/30/25	0.503	0.0	2.6
12/10/25	3.618	1.4	(49.2)	12/18/25	1.915	0.2	(1.3)	12/18/25	3.811	0.0	(15.8)	12/19/25	0.589	(0.9)	11.2
01/28/26	3.487	2.0	(62.3)	02/05/26	1.893	0.3	(3.5)	02/05/26	3.665	1.3	(30.4)	01/23/26	0.692	0.1	21.5
03/18/26	3.377	2.5	(73.3)	03/19/26	1.854	0.3	(7.4)	03/19/26	3.608	1.0	(36.1)				
04/29/26	3.313	2.5	(79.8)	04/30/26	1.840	(0.1)	(8.8)	04/30/26	3.494	0.7	(47.5)				
06/17/26	3.181	2.3	(92.9)	06/11/26	1.813	0.7	(11.5)	06/18/26	3.465	1.2	(50.4)				

Source: Morgan Stanley Research, Bloomberg

Economic Releases

Developed Markets

- **Germany October Ifo Business Climate:** 88.4 (MSe: 88.0; C: 88.0; P: 87.7)

Emerging Markets

- **Thailand September Customs Trade Balance:** $1.275bn (C: -$199mn; P: -$1.964bn); **Customs Exports:** 19.0% y/y (C: 7.2%; P: 5.8%) & **Customs Imports:** 17.2% y/y (C: 10.4%; P: 15.8%)
- **Taiwan September Unemployment Rate:** 3.35% (MSe: 3.40%; C: 3.40%; P: 3.35%)
- **Hong Kong September Trade Balance:** -HK$50.2bn (C: -HK$49.5bn;

P: -HK$25.4bn); **Exports:** 16.1% y/y (C: 15.4%; P: 14.5%) & **Imports:** 13.6% y/y (C: 13.5%; P: 11.5%)

- **Turkey September Unemployment Rate:** 8.6% (P: 8.5%)
- **Turkey October Capacity Utilization:** 74.2% (P: 74.0%)
- **Israel September Unemployment Rate:** 3.0% (P: 2.9%)
- **Israel September Trade Balance:** -USD3424mn (P: -USD3366)

The Day Ahead

Developed Markets

- Mon 8:00pm EST - **UK October BRC Shop Price Index** (MSe: 1.5% y/y, 1.6%, 1.4%). We assume some softening in core goods and food pressures to be captured by the BRC index, which tends to be sampled early in the month
- Tue 7:00am BST - **Germany November GfK Consumer Confidence** (MS: -22.1; C: -22; P: -22.3)
- 9:30am BST - **ECB Governing Council member Panetta** will speak in Rome
- 9:00am EST - **US August S&P Cotality CS 20-City** (MSe: 3.1% m/m ; C: -0.1%; P: -0.1%). In the S&P/Case-Shiller home price indexes, monthly prices have been declining since the beginning of the year. As of July, the annual increase slowed to 1.8%, down from 5.9% in the same month last year.
- 10:00am EST - **US October Conference Board Consumer Confidence** (C: 93.4; P: 94.2). The labor market differential has declined this year, reaching 7.8% in September. While this is low compared to recent years, it remains above levels seen during parts of previous expansions. Consumer confidence has been relatively stable since May, though it fell by 3.6 points to 94.2 in September. For reference, the average in 2024 was 104.5.

Emerging Markets

- Tue 5:00pm EST - **Chile BCCh Monetary Policy Decision** (MSe: 4.75%; P: 4.75%; C: 4.75%) We expect BCCh to hold rates. Persistent core inflation supports a cautious stance, while near-neutral monetary conditions and stronger domestic demand reduce urgency for further easing. Markets are pricing ~13bps of cuts for December and ~22bps from January to October 2026, with a terminal rate of ~4.40%. Read more

- 7:00am HKT - **Korea 3Q25 Advance GDP** (MSe: 1.7% y/y; C: 1.5% y/y; P: 0.6% y/y)
- 6:30pm HKT - **India September Industrial Production** (C: 2.9% y/y; P: 4.0%)
- 8:00am EST - **Mexico September Unemployment Rate NSA** (P: 2.93%; C: 2.86%) Consensus expects a modest decline, reinforcing labor market resilience. Despite this strength, Banxico easing remains priced in, with ~41bps of cuts through year-end and ~32bps in 2026, implying a terminal rate near 6.77% by March.

For more details, please refer to our economists' weekly reports and data previews:

- *US*
- *Japan*
- *Europe*
- *Asia/Pacific*
- *LatAm*
- *CEEMEA*

Please click here if you would like to receive our weekly strategic advice on global rates, FX, and EM. See our latest Global Macro Strategist: It Ain't What You Don't Know That Gets You Into Trouble... and recent publications:

- Argentina Economics, Sovereign Credit Strategy & Equity Strategy: Election Reaction - A Landslide Makes Reform Momentum Sustainable
- G10 FX Strategy: The Dollar and its Risks
- G10 FX Strategy: Long Scandies Versus EUR

G4 Smarter (Beta) Trading Partners

MORGAN STANLEY & CO. LLC

Matthew Hornbach, CMT
Matthew.Hornbach@morganstanley.com
+1 212 761-1837

Since 2000, enhancements to a G4 10y government bond futures momentum strategy have produced higher Sharpe ratios and stronger returns relative to total return government bond indices for the G4: US, Germany, Japan, and the UK. See A "Smarter" (Beta) Way to Trade G4 10y Futures Duration? for more information on these strategies.

Exhibit 5: Trading Signals for G4 Smarter (beta) Trading Strategy

Current Risk, G4 10y Futures	G4 Strategy Weight	Trade Longs Portfolio	Fade Shorts Portfolio	Total Risk Trade Longs Only	Total Risk Trade Longs/Fade Shorts (max 200%)	Trade Longs Portfolio Entry Date	Trade Longs Portfolio Exit Date	Fade Shorts Portfolio Entry Date	Fade Shorts Portfolio Exit Date
JB 10y Future	32.50%	100%	100%	100%	200%	10/15/2025	11/20/2025	9/19/2025	10/28/2025
GE 10y Future	29.25%	100%	0%	100%	100%	10/7/2025	11/11/2025	-	-
US 10y Future	30.50%	100%	100%	100%	200%	10/14/2025	11/19/2025	9/24/2025	10/30/2025
UK 10y Future	7.75%	100%	100%	100%	200%	10/15/2025	11/19/2025	9/25/2025	10/30/2025

Source: Morgan Stanley Research

Trading Strategy 1 – "Trade Longs/Fade Shorts"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, buy the futures contract and hold for a 25-business-day period. In short, this strategy buys futures when the Simple Moving Average Crossover (SMAX) generates both a long and a short signal, given the historical outperformance of long signals traded long and the underperformance of short signals traded short. Given that the SMAX could generate both a long and a short signal within the predefined holding period, an investor may have a 200% long position since each of the two signals would be traded in separate portfolio sleeves.

Trading Strategy 2 – Trade "Longs Only"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, do nothing. In short, an investor *only* trades long signals initiated by the SMAX given their historical precedent to outperform.

US Holiday Trading Strategy

Currently Active Strategies

None.

Combining the individual trading signals based on 10y Treasury future (TY) price patterns before and after US holidays has generated an annual strategy with no down years since 1987. See Trading 10y Treasury Futures Around US Holidays for more information.

Using historical data since 1983, we created a trading strategy using 10y Treasury futures (TY) that trades 11 times per year, either long or short, before or after the following US holidays: New Year’s Day, Martin Luther King Day, Presidents' Day, Good Friday, Memorial Day, Independence Day, Labor Day, Columbus Day, Thanksgiving, and Christmas. Exhibit 6 displays the trading rules:

Exhibit 6: Trading rules for US holiday trading model using US 10y futures contracts (TY)

Holiday	Day of Week	Direction	Before/After Holiday	Trading Rule (business days, 3pm NY unless specified)	Trade Entry (business days)	Trade Exit (business days)
New Year's Day	Rotates	Long TY	After	Buy TY New Years +1, Sell TY New Years +4	New Year + 1	New Year + 4
Martin Luther King Day 1	Monday	Long TY	Before	Buy TY Tuesday before MLK, Sell Friday before MLK	MLK - 4	MLK - 1
Martin Luther King Day 2	Monday	Short TY	After	Sell TY Tuesday after MLK, Buy Friday after MLK	MLK + 1	MLK + 4
President's Day	Monday	Long TY	After	Buy TY Tuesday after Pres. Day, Sell TY Friday after Pres. Day	Pres. Day + 1	Pres. Day + 4
Good Friday	Friday	Long TY	After	Buy TY Monday after Good Friday, Sell TY Friday after Good Friday	Good Friday + 1	Good Friday + 5
Memorial Day	Monday	Long TY	After	Buy TY Tuesday after Mem. Day, Sell TY Friday after Mem. Day	Mem. Day + 1	Mem. Day + 4
Independence Day (July 4th)	Rotates	Long TY	After	Buy TY Independence Day + 2, Sell TY Independence Day + 4	Ind. Day + 2	Ind. Day + 4
Labor Day	Monday	Long TY	Before	Buy TY Monday before Labor Day, Sell TY Thursday before Labor Day	Labor Day - 4	Labor Day - 2
Columbus Day	Monday	Short TY	Before	Sell TY Monday before Col. Day, Buy TY Thursday before Col. Day	Colum. Day - 5	Colum. Day - 2
Thanksgiving	Thursday	Long TY	Before	Buy TY Friday before Thanksgiving, Sell TY Tuesday before Thanksgiving	Thanksgiving - 4	Thanksgiving - 2
Christmas	Rotates	Short TY	Before	Sell TY Christmas Day - 4, Buy TY Christmas Day - 1	Christmas - 4	Christmas - 1

Source: Morgan Stanley Research. Note: Past performance is no guarantee of future results.

Global Macro Strategy Team

MORGAN STANLEY & CO. LLC	**Matthew Hornbach, CMT** Matthew.Hornbach@morganstanley.com	Global Head of Macro Strategy	+1 212 761-1837
	Bradley Tian	Global Macro Commentary	+1 212 761-0826
	Martin Tobias, CFA, CMT	US Rates Strategist	+1 212 761-6076
	Shaun Zhou	US Rates Strategist	+1 212 761-3348
	Aryaman Singh	US Rates Strategist	+1 212 761-1993
	Eli Carter	US Rates Strategist	+1 212 761-4703
MORGAN STANLEY & CO. LLC	**Andrew Watrous**	G10 FX Strategist	+1 212 761-5287
	Molly Nickolin	G10 FX Strategist	+1 212 761-3592
	Simon Waever Simon.Waever@morganstanley.com	Head of EM Sovereign Credit and Latin America Fixed Income Strategy	+1 212 296-8101
	Ioana Zamfir	Latin America Macro Strategist	+1 212 761-4012
	Emma Cerda	Latin America Sovereign Credit	+1 212 761-2344
MORGAN STANLEY & CO. INTERNATIONAL PLC	**James K. Lord** James.Lord@morganstanley.com	Global Head of FXEM Strategy	+44 20 7677-3254
	Gianluca Salford Luca.Salford@morganstanley.com	Head of European Rates Strategy	+44 20 7677-1337
	Lorenzo Testa	Euro Area Rates Strategist	+44 20 7677-0337
	Fabio Bassanin, CFA	UK Rates Strategist	+44 20 7425-1869
	Maria Chiara Russo	Euro Area Rates Strategist	+44 20 7677-3499
	David S. Adams, CFA David.S.Adams@morganstanley.com	Head of G10 FX Strategy	+44 20 7425-3518
	Neville Mandimika	CEEMEA Sovereign Credit Strategist	+44 20 7425-2509
	Arnav Gupta	CEEMEA Rates Strategist	+44 20 7677-0382
	David Cueva Oemer	CEEMEA Rates Strategist	+44 20 7425-2143
MORGAN STANLEY ASIA LIMITED+	**Gek Teng Khoo**	AXJ Macro Strategist	+852 3963-0303
MORGAN STANLEY MUFG SECURITIES CO., LTD.	**Koichi Sugisaki** Koichi.Sugisaki@morganstanleymufg.com	Head of Japan Macro Strategy	+81 3 6836-8428
	Hiromu Uezato	Japan Macro Strategist	+81 3 6836-8431

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/researchdisclosures, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: David S. Adams, CFA; Fabio Bassanin, CFA; Eli P Carter; David Cueva; Arnav Gupta; Matthew Hornbach; Gek Teng Khoo; James K Lord; Molly Nickolin; Sofia Palacios; Nimish M Prabhune; Maria Chiara Russo; Luca Salford; Aryaman Singh; Koichi Sugisaki; Lorenzo Testa; Bradley Tian; Martin W Tobias, CFA; Hiromu Uezato; Andrew M Watrous; Ioana Zamfir; Shaun Zhou.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Germany.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Germany.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Germany.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Germany.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2025)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1499	41%	389	46%	26%	702	41%
Equal-weight/Hold	1618	44%	375	44%	23%	782	45%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	577	16%	88	10%	15%	234	14%
Total	3,698		853			1719	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding the relationship between the companies that are the subject of Morgan Stanley Research and Morgan Stanley Smith Barney LLC or Morgan Stanley or any of their affiliates, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to www.morganstanley.com/researchdisclosures.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley & Co. International PLC and its affiliates have a significant financial interest in the debt securities of Germany, Japan, United Kingdom.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct